

Near-consistent robust estimations of moments for unimodal distributions

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Resistant to departures, current nonparametric robust estimators are inconsistent with population moments. Here, by constructing invariant mean simultaneously for a semi-parametric distribution and a parametric distribution, near-consistent robust statistics (NRSSs) are proposed for estimating mean and central moments with insignificant asymptotic biases for common continuous unimodal distributions, such as Weibull, gamma, Pareto, lognormal and generalized Gaussian. This article also illuminates the understanding of the common nature of probability distributions and the measures of them.

quantile inequality | invariant | unimodal | adaptive estimation | U-statistics

The asymptotic inconsistencies between sample mean ($\bar{x}$) and nonparametric robust location estimators in asymmetric distributions on the real line have been noticed for more than two centuries (1) but unsolved. Strictly speaking, it is unsolvable because by trimming, some information about the original distribution is removed, making it impossible to estimate the values of the removed parts without distributional assumptions. Newcomb (1886, 1912) provided the first modern approach to this problem by developing a class of estimators that gives "less weight to the more discordant observations" (2, 3). In 1964, Huber (4) used the minimax procedure to obtain M-estimator for contaminated normal distribution, which has played a pre-eminent role in the later development of robust statistics. In previous work on semi-parametric robust mean estimation, although greatly shrinking the biases, binomial mean (BM_ϵ) is still inconsistent for any skewed distribution if $\epsilon > 0$ (if $\epsilon \rightarrow 0$, since the alternating sum of binomial coefficients is zero, $BM \rightarrow \mu$). All robust location estimators commonly used are symmetric due to the universality of the symmetric distributions. One can construct an asymmetric trimmed mean that is consistent for a semi-parametric class of skewed distributions. This approach has been investigated previously, but it is not symmetric and therefore only suitable for some special applications (5). From semi-parametric to parametric, an ideal robust location estimator would have a non-sample-dependent breakdown point (defined in Subsection E) and be consistent with any symmetric distribution and a skewed distribution with finite second moments. This is called an invariant mean. Based on the mean-weighted average-median inequality, the robust mean is defined as

$$rm_{d,\epsilon,n} := \lim_{c \rightarrow \infty} \left(\frac{(WA_{\epsilon,n} + c)^{d+1}}{(\text{median} + c)^d} - c \right),$$

where d is for bias correction, $WA_{\epsilon,n}$ is $BM_{\epsilon,n}$ in the first three Subsections, while other symmetric weighted averages can also be used in practice as long as the inequalities hold. The next two theorems show the significance of this composite estimator.

Theorem .1. *If the second moments are finite, $rm_{d \approx 0.375, \epsilon = \frac{1}{8}}$ is a consistent mean estimator for the exponential and any symmetric distributions and the Pareto distribution with quantile function $Q(p) = x_m(1-p)^{-\frac{1}{\alpha}}$, $x_m > 0$, when $\alpha \rightarrow \infty$.*

Proof. Finding d, ϵ values that make $rm_{d,\epsilon}$ a consistent mean estimator is equivalent to finding the solution of $E[rm_{d,\epsilon}] = E[X]$. Rearranging the definition, $rm_{d,\epsilon} = \lim_{c \rightarrow \infty} \left(\frac{(BM_\epsilon + c)^{d+1}}{(\text{median} + c)^d} - c \right) = (d+1)BM_\epsilon - d\text{median} = \mu$. So, $d = \frac{\mu - BM_\epsilon}{BM_\epsilon - \text{median}}$. The pdf of the exponential distribution is $f(x) = \lambda^{-1}e^{-\lambda^{-1}x}$, $\lambda \geq 0$, $x \geq 0$, the cdf is $F(x) = 1 - e^{-\lambda^{-1}x}$, $x \geq 0$. The quantile function is $Q(p) = \ln\left(\frac{1}{1-p}\right)\lambda$. $E[x] = \lambda$. $E[\text{median}] = Q\left(\frac{1}{2}\right) = \ln 2\lambda$. For the exponential distribution, the expectation of $BM_{\frac{1}{8}}$ is $E[BM_{\frac{1}{8}}] = \lambda \left(1 + \ln\left(\frac{46656}{8575\sqrt{35}}\right)\right)$. Obviously, the scale parameter λ can be canceled out, $d \approx 0.375$. The proof of the second assertion follows directly from the coincidence property. For any symmetric distribution with a finite second moment, $E[BM_\epsilon] = E[\text{median}] = E[X]$. Then $E[rm_{d,\epsilon}] = \lim_{c \rightarrow \infty} \left(\frac{(E[X] + c)^{d+1}}{(E[X] + c)^d} - c \right) = E[X]$. For the Pareto distribution, the proof is more general. The mean for the Pareto distribution is $\frac{\alpha x_m}{\alpha - 1}$. The d value with two unknown percentiles p_1 and p_2 for the Pareto distribution is $d_{Pareto} = \frac{\frac{\alpha x_m}{\alpha - 1} - x_m(1-p_1)^{-\frac{1}{\alpha}}}{x_m(1-p_1)^{-\frac{1}{\alpha}} - x_m(1-p_2)^{-\frac{1}{\alpha}}}$. Since any weighted average can be expressed as an integral of the quantile function, $\lim_{\alpha \rightarrow \infty} \frac{\frac{\alpha}{\alpha-1} - (1-p_1)^{-1/\alpha}}{(1-p_1)^{-1/\alpha} - (1-p_2)^{-1/\alpha}} = -\frac{\ln(1-p_1)+1}{\ln(1-p_1) - \ln(1-p_2)}$, the d value for the Pareto distribution when $\alpha \rightarrow \infty$ is always identical to that for the exponential distribution, regardless of the kind of weighted average used. $\square$

Theorem .1 implies that for the Weibull, gamma, Pareto, lognormal and generalized Gaussian distribution,

Significance Statement

Bias, variance, and contamination are the three main errors in statistics. Consistent robust estimation is impossible without specific parametric assumptions. Here, based on a paradigm shift inspired by mean-median-mode inequality, Bickel-Lehmann spread, M-estimator, and adaptive estimation, near-consistent robust statistics are proposed for unimodal distributions, while the variances are often smaller than those of the sample moments.

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$rm_{d \approx 0.375, \epsilon = \frac{1}{8}}$ is consistent for at least one particular case within these two-parameter distributions. The biases of $rm_{d \approx 0.375, \epsilon = \frac{1}{8}}$ for distributions with skewness between those of the exponential and symmetric distributions are tiny (SI Dataset S1). Therefore, $rm_{d \approx 0.375, \epsilon = \frac{1}{8}}$ has excellent performance for all these common unimodal distributions (SI Dataset S1).

Besides introducing the concept of invariant mean, the purpose of this paper is to demonstrate that the estimation of central moments can be transformed into a location estimation problem by using U -statistics, the U -statistic structure has nice properties, and, in light of previous works, a series of sophisticated yet efficient robust estimators can be constructed whose biases are typically smaller than the variances ($n = 5400$, Table 1) for unimodal distributions.

Background and Main Results

A. Invariant mean. It has long been known that a theoretical model can be adjusted to fit the first two moments of the observed data. A continuous distribution belonging to a location-scale family has the form $F(x) = F_0\left(\frac{x-\mu}{\lambda}\right)$, where F_0 is a "standard" distribution. Then, $\frac{x-\mu}{\lambda} = Q_0(p) \rightarrow x = \lambda Q_0(p) + \mu$. Since $F(x) = Q^{-1}(x) \rightarrow x = Q(p) = \lambda Q_0(p) + \mu$. So, any weighted average can be expressed as $\lambda W_{estimator}(\epsilon) + \mu$, where $W_{estimator}(\epsilon)$ is a function of $Q_0(p)$ according to the definition of the weighted average. The simultaneous cancellation of μ and λ in $\frac{(\lambda W_{\mu}(\epsilon) + \mu) - (\lambda W_{BM_{\epsilon}}(\epsilon) + \mu)}{(\lambda W_{BM_{\epsilon}}(\epsilon) + \mu) - (\lambda W_{median}(\epsilon) + \mu)}$ ensures that d is a constant. Consequently, the roles of BM_{ϵ} and median in $rm_{d,b}$ can be replaced by any weighted averages.

The performance in heavy-tailed distributions can be improved further by constructing the quantile mean as

$$qm_{d,\epsilon,n} := \hat{Q}_n \left(\left(\hat{F}_n(WA_{\epsilon,n}) - \frac{1}{2} \right) d + \hat{F}_n(WA_{\epsilon,n}) \right),$$

provided that $\hat{F}_n(WA_{\epsilon,n}) \geq \frac{1}{2}$, where $\hat{F}_n(x)$ is the empirical cumulative distribution function of the sample, $\hat{Q}_n$ is the sample quantile function. The most popular method for computing the sample quantile function was proposed by Hyndman and Fan in 1996 (6). To minimize the finite sample bias, here, $\hat{F}_n(x) := \frac{1}{n} \left(\frac{x - \hat{Q}_n(\frac{sp}{n})}{\hat{Q}_n(\frac{1}{n}(sp+1)) - \hat{Q}_n(\frac{sp}{n})} + sp \right)$, where $sp = \sum_{i=1}^n \mathbf{1}_{X_i \leq x}$, $\mathbf{1}_A$ is the indicator of event A . The solution of $\hat{F}_n(WA_{\epsilon,n}) < \frac{1}{2}$ is reversing the percentile by $1 - \hat{F}_n(WA_{\epsilon,n})$, the obtained percentile is also reversed*. Moreover, in extreme heavy-tailed distributions, the calculated percentile can exceed the breakdown point of WA_{ϵ} , so the percentile will be modified to $1 - \epsilon$ if this happens. The quantile mean uses the location-scale invariant in a different way, as shown in the following proof.

Theorem A.1. $qm_{d \approx 0.321, \epsilon = \frac{1}{8}}$ is a consistent mean estimator for the exponential, Pareto ($\alpha \rightarrow \infty$) and any symmetric distributions provided that the second moments are finite.

Proof. Similarly, rearranging the definition, $d = \frac{F(\mu) - F(BM_{\epsilon})}{F(BM_{\epsilon}) - \frac{1}{2}}$.

Recall the cdf is $F(x) = 1 - e^{-\lambda^{-1}x}$, $x \geq 0$, the expectation

*Without loss of generality, in the following discussion, only the $\hat{F}_n(WA_{\epsilon,n}) \geq \frac{1}{2}$ case will be considered.

of BM_{ϵ} can be expressed as $\lambda W_{BM_{\epsilon}}(\epsilon)$, so $F(BM_{\epsilon})$ is free of λ . When $\epsilon = \frac{1}{8}$, $d = \frac{-e^{-1} + e^{-\left(1 + \ln\left(\frac{46656}{8575\sqrt{35}}\right)\right)}}{\frac{1}{2} - e^{-\left(1 + \ln\left(\frac{46656}{8575\sqrt{35}}\right)\right)}} \approx 0.321$.

The proof of the symmetric case is similar. Since for any symmetric distribution with a finite second moment, $F(E[BM_{\epsilon}]) = F(\mu) = \frac{1}{2}$. Then, the expectation of the quantile mean is $qm_{d,\epsilon} = F^{-1}\left((F(\mu) - \frac{1}{2})d + F(\mu)\right) = F^{-1}\left(0 + F(\mu)\right) = \mu$.

For the assertion related to the Pareto distribution, the cdf of it is $1 - \left(\frac{x_m}{x}\right)^{\alpha}$. So, the d value with two unknown percentile p_1 and p_2 is

$$d_{Pareto} = \frac{1 - \left(\frac{x_m}{\frac{x_m}{\alpha-1}}\right)^{\alpha} - \left(1 - \left(\frac{x_m}{x_m(1-p_1)^{-\frac{1}{\alpha}}}\right)^{\alpha}\right)}{\left(1 - \left(\frac{x_m}{x_m(1-p_1)^{-\frac{1}{\alpha}}}\right)^{\alpha}\right) - \left(1 - \left(\frac{x_m}{x_m(1-p_2)^{-\frac{1}{\alpha}}}\right)^{\alpha}\right)} = \frac{1 - \left(\frac{\alpha-1}{\alpha}\right)^{\alpha-p_1}}{\frac{1-p_1}{p_1-p_2}}. \text{ When } \alpha \rightarrow \infty, \left(\frac{\alpha-1}{\alpha}\right)^{\alpha} = \frac{1}{e}. \text{ The } d \text{ value for the exponential distribution is identical, since } d_{exp} = \frac{(1-e^{-1}) - \left(1 - e^{-\ln\left(\frac{1}{1-p_1}\right)}\right)}{\left(1 - e^{-\ln\left(\frac{1}{1-p_1}\right)}\right) - \left(1 - e^{-\ln\left(\frac{1}{1-p_2}\right)}\right)} = \frac{1 - \frac{1}{e} - p_1}{p_1 - p_2}. \quad \square$$

The definitions of location and scale parameters are such that they must satisfy $F(x; \lambda, \mu) = F\left(\frac{x-\mu}{\lambda}; 1, 0\right)$. Recall that $x = \lambda Q_0(p) + \mu$, so the percentile of any weighted average is free of λ and μ , guaranteeing the validity of the quantile mean. $qm_{d \approx 0.321, \epsilon = \frac{1}{8}}$ works better in the fat-tail scenarios (SI Dataset S1). Theorem .1 and A.1 show that $rm_{d \approx 0.375, \epsilon = \frac{1}{8}}$ and $qm_{d \approx 0.321, \epsilon = \frac{1}{8}}$ are both consistent mean estimators for any symmetric distribution and a skewed distribution with finite second moments. It's obvious that the breakdown points of $rm_{d \approx 0.375, \epsilon = \frac{1}{8}}$ and $qm_{d \approx 0.321, \epsilon = \frac{1}{8}}$ are both $\frac{1}{8}$. Therefore they are all invariant means.

It is constructive to consider a symmetric weighted average as a mixture of trimmed mean and symmetric quantile average. Since the quantile inequality is not a necessary condition of the trimming inequality, the safest choice of symmetric weighted average in rm and qm is the $\frac{1}{4}$ -trimmed mean (SI Dataset S1). Using a less-biased symmetric weighted average can generally improve the performance, but the risk of violation is higher in practice. However, another key factor determining the risk of violation is the skewness of the distribution. Suppose there is a near-symmetric distribution S such that SQA_{ϵ} as a function of ϵ is monotonic increasing from 0 to u and monotonic decreasing from u to $\frac{1}{2}$, and $\mu = m$. S does not follow the quantile inequality according to the definition. Depending on the degree of fluctuation, $|SQA_u - m|$, the mean-SWA $_{\epsilon}$ -median inequality in most cases will not be valid for S . Then, the monotonic increase from 0 to u implies that $Q'(\epsilon) \geq Q'(1-\epsilon) \Leftrightarrow f(Q(1-\epsilon)) \geq f(Q(\epsilon))$ always holds for $0 \leq \epsilon \leq u$. Similarly, $Q'(\epsilon) \leq Q'(1-\epsilon) \Leftrightarrow f(Q(\epsilon)) \geq f(Q(1-\epsilon))$ always holds for $u \leq \epsilon \leq \frac{1}{2}$. Transforming S with $\chi(x)$ will decrease $f(Q(\epsilon))$, and the decrease rate, due to the order, is much smaller than $f(Q(1-\epsilon))$. That means, as the second derivative of $\chi(x)$ increases, eventually, after a point, $f(Q(\epsilon))$ will always be greater than $f(Q(1-\epsilon))$, i.e., the SQA_{ϵ} function will be monotonic decreasing and S will eventually follow the quantile inequality. $|\mu - m|$ will increase so that the skewness will increase. Even the transformation is not enough to make

the SQA function strictly monotonic; suppose it is generally decreasing in $[0, u]$, but increasing in $[u, \frac{1}{2}]$, since $1 - 2\epsilon$ of the symmetric quantile averages will be included in the computation of BM_ϵ , as long as $|u - \frac{1}{2}| \ll 1 - 2\epsilon$, and other parts of the SQA function satisfy the inequality constraints which define the ν th quantile inequality, the mean- BM_ϵ -median inequality will still be valid (as an example, the SQA function is non-monotonic when the shape parameter of the Weibull distribution $\alpha > \frac{1}{1-\ln(2)} \approx 3.259$ as shown in the previous article, yet the mean- $\text{BM}_{\frac{1}{8}}$ -median inequality is still valid when $\alpha \leq 3.322$). Accordingly, in a family of distributions that differ by a skewness-increasing transformation, the violation of the mean- WA_ϵ -median inequality often only occurs when the distribution is near-symmetric, but the over-corrections in rm and qm are dependent on the WA_ϵ -median difference, which is correlated to the skewness, so the over-correction, if it happens, is often tiny with a moderate d . This provides a more general perspective, in addition to the bias bounds (7), that rm and qm based on the mean- WA_ϵ -median inequality are generally safe.

B. Robust estimations of the central moments. In 1976, Bickel and Lehmann, in their third paper of the landmark series *Descriptive Statistics for Nonparametric Models* (8), generalized a class of estimators called "measures of disperse," which is now often named as Bickel-Lehmann dispersion. As an example, they proposed a first version of the trimmed standard deviation, $\hat{\tau}^2(F; \epsilon) \equiv \tau^2(F; \epsilon)$, for independent and identically distributed random variables X_i with a distribution F , where $\tau^2(F; \epsilon) = \frac{1}{1-2\epsilon} \int_{Q(\epsilon)}^{Q(1-\epsilon)} y dG(y)$, Q is the quantile function of G , G is the distribution of $Y = X^2$. Obviously, when $\epsilon = 0$, the result is equivalent to the second raw moment. In 1979, in the same series (9), they explored another class of estimators called "measures of spread," which "does not require the assumption of symmetry." From that, a popular efficient scale estimator, the Rousseeuw-Croux scale estimator (10), was derived in 1993, but the importance of tackling the symmetry assumption has been greatly underestimated. In the final section of the paper, they considered another two possible versions of the trimmed standard deviations, which were modified symmetrically here for comparison,

$$\left[n \left(\frac{1}{2} - \epsilon \right) \right]^{-\frac{1}{2}} \left[\sum_{k=\frac{n}{2}}^{n(1-\epsilon)} [X_k - X_{n-k+1}]^2 \right]^{\frac{1}{2}}, \quad [1]$$

and

$$\left[\binom{n}{2} (1 - 2\epsilon) \right]^{-\frac{1}{2}} \left[\sum_{k=\binom{n}{2}\epsilon}^{\binom{n}{2}(1-\epsilon)} (X - X')_k^2 \right]^{\frac{1}{2}}, \quad [2]$$

where $(X - X')_1 \leq \dots \leq (X - X')_{\binom{n}{2}}$ are the order statistics of the "pseudo-sample" $X_i - X_j$, $i < j$. The paper ended with, "We do not know a fortiori which of the measures [1] or [2] is preferable and leave these interesting questions open."

Observe that the kernel of the unbiased estimation of the second central moment by using U -statistic is $\psi_2(x_1, x_2) = \frac{1}{2}(x_1 - x_2)^2$. If adding the $\frac{1}{2}$ term in [2], as $\epsilon \rightarrow 0$, the result is the same as the standard deviation estimated by using U -statistic (also pointed by Janssen, Serfling, and Veraverbeke

in 1987) (11). In fact, they also implied that, when ϵ is 0, [2] is $\sqrt{2}$ times the standard deviation.

To address their open questions, the nomenclature used in this paper is introduced as follows:

Nomenclature. Given a robust estimator $\hat{\theta}$. The first part of the name of the robust statistic defined in this paper is the prefix that indicates the type of estimator, while the second part is the name of the population parameter θ , which the estimator is consistent with as $\epsilon \rightarrow 0$. The abbreviation of the estimator is the initial letter(s) of the first part, plus the common abbreviation of the consistent estimator that measures the population parameter. If the estimator is not a U -statistic, the breakdown point, ϵ , is indicated in the subscript of the abbreviation of the estimator. If the estimator is a robustified U -statistic or a composite estimator, the breakdown point of the location estimator is indicated (except the median).

Naturally, the trimmed standard deviation following this nomenclature is $\text{Tsd}_{\epsilon,n} := \left[\text{TM}_\epsilon \left((\psi_2(X_{N_1}, X_{N_2}))_{N=1}^{\binom{n}{2}} \right) \right]^{-\frac{1}{2}}$, where $\text{TM}_\epsilon(Y)$ denotes the ϵ -symmetric trimmed mean with the sequence $(\psi_2(X_{N_1}, X_{N_2}))_{N=1}^{\binom{n}{2}}$ as an input. If the square root is removed, it is named as the trimmed variance ($\text{Tvar}_{\epsilon,n}$). It is now very clear that this definition, essentially the same as [2], should be preferable. Not only because it is essentially a trimmed U -statistic for the standard deviation but also because the quantile inequality of the pseudo-sample distribution is ensured by the next exciting theorem.

Theorem B.1. *The second central moment kernel distribution generated from any continuous unimodal distribution follows the quantile inequality.*

Proof. Let $Q(p)$, $0 \leq p \leq 1$, denote the quantile of the continuous unimodal distribution $f_X(x)$. The corresponding probability density is $f(Q(p))$. Generating the distribution of the pair $(Q(p_i), Q(p_j))$, $i < j$, $p_i < p_j$, the corresponding probability density is $f_{X,X}(Q(p_i), Q(p_j)) = 2f(Q(p_i))f(Q(p_j))$. Transforming the pair $(Q(p_i), Q(p_j))$, $i < j$, by the function $\Phi(x_1, x_2) = x_1 - x_2$, the pairwise difference distribution has a mode that is arbitrary close to $M - M = 0$. The monotonic increasing of the pairwise difference distribution was first implied in its unimodality proof done by Hodges and Lehmann in 1954 (12). Whereas they used induction to get the result, Dharmadhikari and Jogdeo in 1982 (13) gave a modern proof of the unimodality using Khintchine's representation (14). Assuming absolute continuity, Purkayastha (15) introduced a much simpler proof in 1998. Transforming the pairwise difference distribution by squaring and multiplying $\frac{1}{2}$ does not change the monotonicity, making the pdf become monotonically decreasing with mode at zero. In the previous semi-parametric robust mean estimation article, it is proven that a distribution with monotonic pdf always follows the quantile inequality, which gives the desired result. $\square$

Remark. The assumption of continuity of distributions is important for monotonicity because, unlike in the continuous case, it is possible to get pairs with the same value for a discrete distribution. For example, let the probabilities of the singletons $\{1\}$, $\{2\}$, $\{3\}$, $\{4\}$ and $\{5\}$ of a probability mass function of a discrete probability distribution be $\frac{1}{11}$, $\frac{4}{11}$, $\frac{3}{11}$, $\frac{2}{11}$, and $\frac{1}{11}$, respectively. This is a unimodal distribution, but the corresponding ψ_2 distribution is non-monotonic, whose

singletons $\{0\}$, $\{0.5\}$, $\{2\}$, $\{4.5\}$ and $\{8\}$ have probabilities $\frac{21}{66}$, $\frac{24}{66}$, $\frac{2}{14}$, $\frac{6}{66}$, and $\frac{1}{66}$, respectively.

Previously, it was shown that any symmetric distribution with a finite second moment follows the ν th quantile inequality. That means the quantile inequality does not require unimodality, e.g., for a symmetric bimodal distribution, it also follows the quantile inequality. Examples from the Weibull distribution show that unimodality does not guarantee quantile inequality. Theorem B.1 reveals another profound relationship between unimodality and quantile inequality, which is sufficient for trimming inequality.

In 1928, Fisher constructed k -statistics as unbiased estimators of cumulants (16). Halmos (1946) proved that the functional θ admits an unbiased estimator if and only if it is a regular statistical functional of degree k (17). Heffernan (1997) (18) obtained an estimator of the k th central moment by using U -statistics. In 1984, to study generalized L-statistics, Serfling considered the U -statistic structure (19). Gijbels, Janssen and Veraverbeke generalized the trimmed U -statistics in 1988 (20). Due to the combinatorial explosion, the bootstrap (21), introduced by Efron in 1979, is indispensable in large sample studies. In 1981, Bickel and Freedman (22) showed that the bootstrap is asymptotically valid to approximate the original distribution in a wide range of situations, including U -statistics. After that, the limit laws of bootstrapped U -statistics have been intensively studied and proven by Athreya, Ghosh, Low, and Sen (1984) and Helmers, Janssen, and Veraverbeke (1990) (23, 24). Users can check the accuracy by comparing the unbiased central moments (25) to the bootstrap central moments. The weighted k th central moment ($k \leq n$) is defined as,

$$Wkm_{\epsilon,n} := WA_{\epsilon,n} \left((\psi_k(X_{N_1}, \dots, X_{N_k}))_{N=1}^n \right),$$

where $X_{N_1}, \dots, X_{N_k}$ are the n choose k elements from X , $\psi_k(x_1, \dots, x_k) = \sum_{j=0}^{k-2} (-1)^j \left(\frac{1}{k-j} \right) \sum (x_{i_1}^{k-j} \dots x_{i_{j+1}}) + (-1)^{k-1} (k-1) x_1 \dots x_k$, the second summation is over $i_1, \dots, i_{j+1} = 1$ to k with $i_1 < \dots < i_{j+1}$ (9, 16–18). Despite the complexity, the structure of the k th central moment kernel distributions can be elucidated by decomposing.

Theorem B.2. For each pair $(Q(p_i), Q(p_j))$ of the original distribution, let $x_1 = Q(p_i)$ and $x_k = Q(p_j)$, $\Delta = Q(p_i) - Q(p_j)$. The k th central moment kernel distribution, $k > 2$, can be decomposed as a sum of partial pseudodistributions having the interval $(-\left(\frac{k}{3+(-1)^k}\right)^{-1}(-\Delta)^k, \frac{1}{k}(-\Delta)^k)$.

Proof. Similar to Theorem D.1, when generating the distribution of the k -tuple $(Q(p_{i_1}), \dots, Q(p_{i_k}))$, $k > 2$, $i_1 < \dots < i_k$, $p_{i_1} < \dots < p_{i_k}$, the corresponding probability density is $f_{X, \dots, X}(Q(p_{i_1}), \dots, Q(p_{i_k})) = k! f(Q(p_{i_1})) \dots f(Q(p_{i_k}))$. Transforming the distribution of the k -tuple by the function $\psi_k(x_1, \dots, x_k)$, denoting $\bar{\Delta} = \psi_k(Q(p_{i_1}), \dots, Q(p_{i_k}))$. The probability $f_{\Xi_k}(\bar{\Delta}) = \sum_{\bar{\Delta} = \psi_k(Q(p_{i_1}), \dots, Q(p_{i_k}))} f_{X, \dots, X}(Q(p_{i_1}), \dots, Q(p_{i_k}))$ is the sum of the probabilities of all k -tuples such that $\bar{\Delta}$ is equal to $\psi_k(Q(p_{i_1}), \dots, Q(p_{i_k}))$. The following Ξ_k is equivalent.

Ξ_k : Every pair with a difference equal to $\Delta = Q(p_{i_1}) - Q(p_{i_k})$ can generate a partial pseudodistribution (but the integral is not equal to 1, so pseudodistribution) such that $x_2, \dots, x_{k-1}$ exhaust all combinations under the inequality constraints, i.e., $Q(p_{i_1}) = x_1 < x_2 < \dots < x_{k-1} < x_k =$

$Q(p_{i_k})$. The sum of all the partial pseudodistributions with the same Δ is ξ_Δ . The sum of ξ_Δ , i.e., from $\Delta = 0$ to $Q(0) - Q(1)$, is Ξ_k .

The interval of ξ_Δ is the extrema of ψ_k subject to the inequality constraints. Using the Lagrange multiplier, one can easily determine the only critical point at $x_1 = \dots = x_k = 0$, where $\psi_k = 0$. Other candidates are within the boundaries, i.e., $\psi_k(x_1 = x_1, x_2 = x_k, \dots, x_k = x_k)$, $\dots$, $\psi_k(x_1 = x_1, \dots, x_i = x_1, x_{i+1} = x_k, \dots, x_k = x_k)$, $\dots$, $\psi_k(x_1 = x_1, \dots, x_{k-1} = x_1, x_k = x_k)$. $\psi_k(x_1 = x_1, \dots, x_i = x_1, x_{i+1} = x_k, \dots, x_k = x_k)$ can be divided into k groups. If $\frac{k+1-i}{2} \leq j \leq \frac{k-1}{2}$, from $j+1$ st to $k-j$ th group, the g th group has $i \binom{i-1}{g-j-1} \binom{k-i}{j}$ terms having the form $(-1)^{g+1} \frac{1}{k-g+1} x_1^{k-j} x_k^j$, from $k-j+1$ th to $i+j$ th group, the g th group has $i \binom{i-1}{g-j-1} \binom{k-i}{j} + (k-i) \binom{k-i-1}{j-k+g-1} \binom{i}{k-j}$ terms having the form $(-1)^{g+1} \frac{1}{k-g+1} x_1^{k-j} x_k^j$. If $j < \frac{k+1-i}{2}$, from $j+1$ st to $i+j$ th group, the g th group has $i \binom{i-1}{g-j-1} \binom{k-i}{j}$ terms having the form $(-1)^{g+1} \frac{1}{k-g+1} x_1^{k-j} x_k^j$. If $j \geq \frac{k}{2}$, from $k-j+1$ st to j th group, the g th group has $(k-i) \binom{k-i-1}{j-k+g-1} \binom{i}{k-j}$ terms having the form $(-1)^{g+1} \frac{1}{k-g+1} x_1^{k-j} x_k^j$, from $j+1$ th to $j+i$ th group, $i+j < k$, the g th group has $i \binom{i-1}{g-j-1} \binom{k-i}{j} + (k-i) \binom{k-i-1}{j-k+g-1} \binom{i}{k-j}$ terms having the form $(-1)^{g+1} \frac{1}{k-g+1} x_1^{k-j} x_k^j$. The final k th group is the term $(-1)^{k-1} (k-1) x_1^i x_k^{k-i}$. So, if $i+j = k$, $j \geq \frac{k}{2}$, $i \leq \frac{k}{2}$, the summed coefficient of $x_1^i x_k^{k-i}$ is $(-1)^{k-1} (k-1) + \sum_{g=i+1}^{k-1} (-1)^{g+1} \frac{1}{k-g+1} (k-i) \binom{k-i-1}{g-i-1} + \sum_{g=k-i+1}^{k-1} (-1)^{g+1} \frac{1}{k-g+1} i \binom{i-1}{g-k+i-1} = (-1)^{k-1} (k-1) + (-1)^{k+1} + (k-i) (-1)^k + (-1)^k (i-1) = (-1)^{k+1}$. The summation identities are $\sum_{g=i+1}^{k-1} (-1)^{g+1} \frac{1}{k-g+1} (k-i) \binom{k-i-1}{g-i-1} = (k-i) \int_0^1 \sum_{g=i+1}^{k-1} (-1)^{g+1} \binom{k-i-1}{g-i-1} t^{k-g} dt = (k-i) \int_0^1 ((-1)^i (t-1)^{k-i-1} - (-1)^{k+1}) dt = (k-i) \left(\frac{(-1)^k}{i-k} + (-1)^k \right) = (-1)^{k+1} + (k-i) (-1)^k \cdot \sum_{g=k-i+1}^{k-1} (-1)^{g+1} \frac{1}{k-g+1} i \binom{i-1}{g-k+i-1} = \int_0^1 \sum_{g=k-i+1}^{k-1} (-1)^{g+1} i \binom{i-1}{g-k+i-1} t^{k-g} dt = \int_0^1 (i (-1)^{k-i} (t-1)^{i-1} - i (-1)^{k+1}) dt = (-1)^k (i-1)$. If $j < \frac{k+1-i}{2}$, $i > k-1$, if $i = k$, $\psi_k = 0$, if $\frac{k+1-i}{2} \leq j \leq \frac{k-1}{2}$, $\frac{k+1}{2} \leq i \leq k-1$, the summed coefficient of $x_1^i x_k^{k-i}$ is $(-1)^{k-1} (k-1) + \sum_{g=k-i+1}^{k-1} (-1)^{g+1} \frac{1}{k-g+1} i \binom{i-1}{g-k+i-1} + \sum_{g=i+1}^{k-1} (-1)^{g+1} \frac{1}{k-g+1} (k-i) \binom{k-i-1}{g-i-1}$, the same as above. If $i+j < k$, since $\binom{i}{k-j} = 0$, the related terms can be ignored, so, using the binomial theorem and beta function, the summed coefficient of $x_1^{k-j} x_k^j$ is $\sum_{g=j+1}^{i+j} (-1)^{g+1} \frac{1}{k-g+1} i \binom{i-1}{g-j-1} \binom{k-i}{j} = i \binom{k-i}{j} \int_0^1 \sum_{g=j+1}^{i+j} (-1)^{g+1} \binom{i-1}{g-j-1} t^{k-g} dt = \binom{k-i}{j} i \int_0^1 ((-1)^j t^{k-j-1} \left(\frac{t}{t-1} \right)^{1-i}) dt = \binom{k-i}{j} i \frac{(-1)^{j+i+1} \Gamma(i) \Gamma(k-j-i+1)}{\Gamma(k-j+1)} = \frac{(-1)^{j+i+1} i! (k-j-i)! (k-i)!}{(k-j)! j! (k-j-i)!} = (-1)^{j+i+1} \frac{i! (k-i)!}{k!} \frac{k!}{(k-j)! j!}$. The coefficient of $x_1^i x_k^{k-i}$ in $\binom{k}{i}^{-1} (-1)^{1+i} (x_1 - x_k)^k$ is $\binom{k}{i}^{-1} (-1)^{1+i} \binom{k}{i} (-1)^{k-i} = (-1)^{k+1}$, the same as the summed coefficient if $i+j = k$. If $i+j < k$, the coefficient of $x_1^{k-j} x_k^j$ is $\binom{k}{i}^{-1} (-1)^{1+i} \binom{k}{i} (-1)^j$, same as the summed coefficient. Since $\psi_k(x_1 = x_1, \dots, x_i = x_1, x_{i+1} = x_k, \dots, x_k = x_k) =$

$\binom{k}{i}^{-1}(-1)^{1+i}(x_1 - x_k)^k$, the maximum and minimum of ψ_k follow directly from the properties of the binomial coefficient. $\square$

ξ_Δ is closely related to $f_\Xi(\Delta)$, which is the pairwise difference distribution, since the probability density of ξ_Δ is $f_{\Xi_k}(\bar{\Delta}|\Delta)$, $\sum_{\bar{\Delta} = -(\frac{k}{3+(-1)^k})}^{\frac{1}{k}(-\Delta)^k} f_{\Xi_k}(\bar{\Delta}|\Delta) = f_\Xi(\Delta)$. Recall that $f_\Xi(\Delta)$ is monotonic increasing with a mode at the origin if the original distribution is unimodal. So, generally, if ignoring the shape of ξ_Δ , Ξ_k is monotonic left and right around zero. In fact, the median of Ξ_k is also close to zero since it can be cast as a weighted average of the medians of ξ_Δ . When Δ is small, all values of ξ_Δ are close to zero, so the median is arbitrarily close to zero, when Δ is large, the median of ξ_Δ depends on its skewness, but the corresponding weight is much smaller, so even ξ_Δ is highly skewed, the median of Ξ_k will just be slightly dragged away from zero (denote the median of Ξ_k as m_{Ξ_k} , for distributions here, except the generalized Gaussian distribution, $|m_{\Xi_k}|$ s are all $\leq 0.1\sigma$ for Ξ_3 and Ξ_4 , SI Dataset S1). Suppose $m_{\Xi_k} = 0$, for the even ordinal central moment kernel distribution, the average probability density of the left side of zero is greater than that of the right side, since $\frac{\frac{1}{2}}{\binom{k}{2}^{-1}(Q(0)-Q(1))^k} > \frac{\frac{1}{2}}{\binom{k}{k}^{-1}(Q(0)-Q(1))^k}$. That means, on average, the inequality $f(Q(\epsilon)) \geq f(Q(1-\epsilon))$ holds. For the odd ordinal distribution, the discussion is harder since it is generally symmetric. Just consider Ξ_3 , let $x_1 = Q(p_i)$ and $x_3 = Q(p_j)$, changing the value of x_2 from $Q(p_i)$ to $Q(p_j)$ will monotonically change the value of $\psi_3(x_1, x_2, x_3)$, since $\frac{\partial \psi_3(x_1, x_2, x_3)}{\partial x_2} = -\frac{x_1^2}{2} - x_1x_2 + 2x_1x_3 + x_2^2 - x_2x_3 - \frac{x_3^2}{2}$, $-\frac{3}{4}(x_1 - x_3)^2 \leq \frac{\partial \psi_3(x_1, x_2, x_3)}{\partial x_2} \leq -\frac{1}{2}(x_1 - x_3)^2 \leq 0$. If the original distribution is right-skewed, ξ_Δ will be left-skewed, so, for Ξ_3 , the average probability density of the right side of zero will be greater than that of the left side, which means, on average, the inequality $f(Q(\epsilon)) \leq f(Q(1-\epsilon))$ holds (the same result can be inferred from the definition of central moments, the positive of odd order central moment is directly related to the left-skewness of the corresponding kernel distribution). In all, the monotonicity of the pairwise difference distribution guides the general shape of the k th central moment kernel distribution, $k > 2$, forcing it to be unimodal-like with mode and median close to zero, then, the inequality $f(Q(\epsilon)) \leq f(Q(1-\epsilon))$ or $f(Q(\epsilon)) \geq f(Q(1-\epsilon))$ will hold in general. Although the inequality might be violated in a small range, according to the discussion in Subsection A, the mean-SWA_ε-median inequality will still be valid for the central moment kernel distribution.

A key property of the central moment kernel distribution, location invariant, is introduced in the next theorem.

Theorem B.3. $\psi_k(x_1 = \lambda x_1 + \mu, \dots, x_k = \lambda x_k + \mu) = \lambda^k \psi_k(x_1, \dots, x_k)$.

Proof. ψ_k can be divided into k groups. From 1st to $k-1$ th group, the g th group has $\binom{k}{g}\binom{g}{1}$ terms having the form $(-1)^{g+1} \frac{1}{k-g+1} x_1^{k-g+1} \dots x_{i_g}$. The final k th group is the term $(-1)^{k-1}(k-1)x_1 \dots x_k$. Let $x_{i_1} = x_1$, $k \neq g$, the g th group of ψ_k has $\binom{k-l}{g-l}$ terms having the form $(-1)^{g+1} \frac{1}{k-g+1} x_1^{k-g+1} x_2 \dots x_l x_{i_1} \dots x_{i_{g-l}}$, where $x_1, x_2, \dots, x_l$ are fixed, $x_{i_1}, \dots, x_{i_{g-l}}$ are selected such that $i_1, \dots, i_{g-l} \neq 1, 2, \dots, l$. Let $\Psi_k(x_1, x_2, \dots, x_l, x_{i_1}, \dots, x_{i_{g-l}}) =$

$(\lambda x_1 + \mu)^{k-g+1}(\lambda x_2 + \mu) \dots (\lambda x_l + \mu)(\lambda x_{i_1} + \mu) \dots$
 $(\lambda x_{i_{g-l}} + \mu)$, the first group of Ψ_k is $\lambda^k x_1 \dots x_l x_{i_1} \dots x_{i_{g-l}}$,
the h th group of Ψ_k , $h > 1$, has $\binom{k-g+1}{k-h-l+2}$ terms
having the form $\lambda^{k-h+1} \mu^{h-1} x_1^{k-h-l+2} x_2 \dots x_l$. Trans-
forming ψ_k by Ψ_k , then combining all terms with
 $\lambda^{k-h+1} \mu^{h-1} x_1^{k-h-l+2} x_2 \dots x_l$, $x_1^{k-h-l+2} \neq x_1$, the summed
coefficient is $S1_l = \sum_{g=l}^{h+l-1} (-1)^{g+1} \frac{1}{k-g+1} \binom{k-g+1}{k-h-l+2} \binom{k-l}{g-l} =$
 $\sum_{g=l}^{h+l-1} (-1)^{g+1} \frac{(k-l)!}{(h+l-g-1)!(k-h-l+2)!(g-l)!} = 0$, since the
summation is starting from l , ending at $h+l-1$, the first
term includes the factor $g-l=0$, the final term includes
the factor $h+l-g-1=0$, the terms in the middle are
also zero due to the factorial property. Another possible
choice is letting one of $x_{i_2} \dots x_{i_g}$ equal to x_1 , the g th
group of ψ_k has $(k-h) \binom{h-1}{g-k+h-1}$ terms having the form
 $(-1)^{g+1} \frac{1}{k-g+1} x_1 x_2 \dots x_j^{k-g+1} \dots x_{k-h+1} x_{i_1} \dots x_{i_{g-k+h-1}}$,
provided that $k \neq g$, $2 \leq j \leq k-h+1$, where
 $x_1, \dots, x_{k-h+1}$ are fixed, x_j^{k-g+1} and $x_{i_1}, \dots, x_{i_{g-k+h-1}}$
are selected. Transforming these terms by
 $\Psi_k(x_1, x_2, \dots, x_j, \dots, x_{k-h+1}, x_{i_1}, \dots, x_{i_{g-k+h-1}}) =$
 $(\lambda x_1 + \mu)(\lambda x_2 + \mu) \dots (\lambda x_j + \mu)^{k-g+1} \dots (\lambda x_{k-h+1} + \mu)$
 $(\lambda x_{i_1} + \mu) \dots (\lambda x_{i_{g-k+h-1}} + \mu)$, then, there
are $k-g+1$ terms having the form
 $\lambda^{k-h+1} \mu^{h-1} x_1 x_2 \dots x_{k-h+1}$. So, the combined result
is $(-1)^{g+1} (k-h) \binom{h-1}{g-k+h-1} \lambda^{k-h+1} \mu^{h-1} x_1 x_2 \dots x_{k-h+1}$.
Transforming the final k th group of ψ_k by
 Ψ_k , then, there is one term having the form
 $(-1)^{k-1} (k-1) \lambda^{k-h+1} \mu^{h-1} x_1 x_2 \dots x_{k-h+1}$. Another
possible combination is that the g th group of ψ_k con-
tains $(g-k+h-1) \binom{h-1}{g-k+h-1}$ terms having the form
 $(-1)^{g+1} \frac{1}{k-g+1} x_1 x_2 \dots x_{k-h+1} x_{i_1} \dots x_{i_{g-k+h-1}}$,
there is only one term having the form
 $\lambda^{k-h+1} \mu^{h-1} x_1 x_2 \dots x_{k-h+1}$. The above summation $S1_l$
should also be included, i.e., $x_1^{k-h-l+2} = x_1$, $k = h+l-1$, so,
combining all terms with $\lambda^{k-h+1} \mu^{h-1} x_1 x_2 \dots x_{k-h+1}$, according
to the binomial theorem, the summed coefficient is $S2_l =$
 $\sum_{g=k-h+1}^{k-1} (-1)^{g+1} \binom{h-1}{g-k+h-1} (k-h+1 + \frac{g-k+h-1}{k-g+1}) +$
 $(-1)^{k-1} (k-1) = (-1)^k + (-1)^k (k-h) + (h-2)(-1)^k +$
 $(-1)^{k-1} (k-1) = 0$. The result is the same if replacing x_1
with x_i , where i is from 2 to k , and replacing x_l with other
 x_i . Thus, all terms including μ can be canceled out. The
proof is complete by noticing that the remaining part is
 $\lambda^k \psi_k(x_1, \dots, x_k)$. $\square$

Consider two continuous distributions belonging to the same location-scale family, their corresponding k th central moment kernel distributions only differ in scaling. So d is invariant, as shown above. The robust k th central moment, based on rm , is defined by,

$$rkm_{d,\epsilon,n} := (d+1) Wkm_{\epsilon,n} - d mkm_{\epsilon,n},$$

where $Wkm_{\epsilon,n}$ is using the binomial k th central moment ($Bkm_{\epsilon,n}$) here, $mkm_{\epsilon,n}$ is the median k th central moment. Similarly, the quantile will not change after scaling. The quantile k th central moment is thus defined as

$$qkm_{d,\epsilon,n} := \hat{Q}_n \left(\left(pWkm - \frac{1}{2} \right) d + pWkm \right),$$

where $pWkm = \hat{F}_n(Wkm_{\epsilon,n})$, $\hat{F}_n$ is the empirical cumulative distribution function of the corresponding central moment kernel distribution.

Table 1. The performance of invariant moments for 5 common unimodal distributions compared to current popular methods

Errors	TM $\frac{1}{s}$	H-L	WM $\frac{1}{s}$	OSH	BM $\frac{1}{s}$	rm $\frac{1}{s}$	qm $\frac{1}{s}$	Tsd $\frac{1}{s}$	rvar $\frac{1}{s}$	qvar $\frac{1}{s}$	rtm $\frac{1}{s}$	qtm $\frac{1}{s}$	rfm $\frac{1}{s}$	qfm $\frac{1}{s}$
WAAB	0.128	0.109	0.078	0.052	0.057	0.002	0.004	0.237	0.031	0.013	0.025	0.009	0.071	0.018
WAB $_{n=5400}$	0.128	0.108	0.078	0.052	0.057	0.002	0.005	0.235	0.031	0.013	0.025	0.009	0.070	0.019
WSE $\vee$ WSSE	0.014	0.014	0.014	0.015	0.015	0.016	0.017	0.015	0.016	0.018	0.012	0.014	0.012	0.016
WAAB	0.128	0.109	0.078	0.052	0.057	0.003	0.004	0.237	0.007	0.007	0.006	0.005	0.019	0.014
WAB $_{n=5400}$	0.128	0.108	0.078	0.052	0.057	0.003	0.004	0.235	0.006	0.007	0.007	0.006	0.017	0.015
WSE $\vee$ WSSE	0.014	0.014	0.014	0.015	0.015	0.016	0.017	0.015	0.018	0.018	0.016	0.015	0.019	0.017

Errors	m	BWM $\frac{1}{s}$	SQM $\frac{1}{s}$	SM $\frac{1}{g}$	$im_{b,\frac{1}{s}}$	$im_{v,\frac{1}{s}}$	$ivar_{b,\frac{1}{s}}$	$ivar_{v,\frac{1}{s}}$	$itm_{b,\frac{1}{s}}$	$itm_{v,\frac{1}{s}}$	$ifm_{b,\frac{1}{s}}$	$ifm_{v,\frac{1}{s}}$
WAAB	0.205	0.104	0.057	0.070	0.004	0.003	0.007	0.008	0.006	0.007	0.011	0.023
WAB $_{n=5400}$	0.205	0.104	0.057	0.070	0.004	0.003	0.007	0.007	0.006	0.007	0.009	0.019
WSE $\vee$ WSSE	0.016	0.014	0.014	0.014	0.017	0.017	0.019	0.018	0.016	0.016	0.021	0.017

Errors	$\bar{x}$	im_b	im_v	var	var $_{bs}$	ivar $_b$	ivar $_v$	tm	tm $_{bs}$	itm $_b$	itm $_v$	fm	fm $_{bs}$	ifm $_b$	ifm $_v$
AAB	0.000	0.001	0.001	0.000	0.000	0.002	0.006	0.000	0.000	0.004	0.009	0.000	0.000	0.005	0.015
RMSE	0.014	0.018	0.017	0.017	0.017	0.019	0.019	0.020	0.019	0.020	0.020	0.026	0.023	0.022	0.024
AB $_{n=5400}$	0.000	0.001	0.001	0.000	0.001	0.002	0.005	0.001	0.004	0.002	0.008	0.001	0.009	0.004	0.013
SSE	0.014	0.018	0.017	0.017	0.017	0.019	0.018	0.020	0.018	0.019	0.018	0.024	0.021	0.021	0.017

The first section uses the exponential distribution as the consistent distribution, while the second section uses the Weibull distribution ($rkurt$ or $qkurt$ is used for reforming the ds starting from the exponential distribution). $BM\frac{1}{s}$ is the weighted average used in robust/quantile moments. The third section uses the Weibull distribution plus optimization (ikm_v is invariant k th moment, variance-optimized, ikm_b is invariant k th moment, bias-optimized). The fourth section uses the Weibull distribution for the Weibull distribution; the breakdown points are all $\frac{1}{5}$ (not indicated). The average asymptotic bias (AAB, $n \rightarrow \infty$), root mean square error (RMSE, $n = 5400$), average bias (AB, $n = 5400$) and variance (SE $\vee$ SSE, $n = 5400$) of the estimators are in the units of the standard deviations of the kernel distributions (Methods). bs indicates the bootstrap moments. W means that the results were weighted by the number of Google Scholar search results on May 30, 2022 (including synonyms). The calibrations of d values and the computations of AAB, AB, and SSE were described in Subsection D and Methods. The detailed results and related codes are available in SI Dataset S1 and [GitHub](#).

Finally, for standardized moments, quantile skewness and quantile kurtosis are defined to be $qskew_{d,\epsilon,n} := \frac{qtm_{d,\epsilon,n}}{qsd_{d,\epsilon,n}^3}$ and $qkurt_{d,\epsilon,n} := \frac{qfm_{d,\epsilon,n}}{qsd_{d,\epsilon,n}^4}$. Quantile standard deviation ($qsd_{d,\epsilon,n}$), robust standard deviation ($rsd_{d,\epsilon,n}$), quantile third central moment ($qtm_{d,\epsilon,n}$), quantile fourth central moment ($qfm_{d,\epsilon,n}$), robust third central moment ($rtm_{d,\epsilon,n}$), robust fourth central moment ($rfm_{d,\epsilon,n}$), robust skewness ($rskew_{d,\epsilon,n}$), and robust kurtosis ($rkurt_{d,\epsilon,n}$) are all defined similarly as above and not repeated here. The transformation to a location problem can also empower related statistical tests. From the better performance of the quantile mean in heavy-tailed distributions, quantile central moments are much better than robust central moments regarding asymptotic bias.

To avoid confusion, the robust location estimations of the kernel distributions here are very different from Joly and Lugosi (2016) and Laforgue, Clemencon, and Bertail (2019)'s approach (26, 27), which is computing the median of all U -statistics from different blocks based on the median of means technique (28–34).

C. A two-parameter distribution as the consistent distribution. Up to this point, the consistent robust estimation has been limited to a parametric location-scale distribution. In practice, the location parameter is often omitted for simplicity. In 1894, Pearson (35) introduced a family of continuous probability distributions, which are now often characterized by the square of the skewness and the kurtosis. Following that, a distribution specified by a shape parameter (denoted as α here) and a scale parameter (denoted as λ here) is often referred to as a two-parameter distribution. Weibull, gamma,

Pareto, lognormal, and generalized Gaussian distributions are all two-parameter unimodal distributions. α can be converted to skewness or kurtosis, e.g., for the gamma distribution, the skewness is $\frac{2}{\sqrt{\alpha}}$, the kurtosis is $\frac{6}{\alpha} + 3$. If α is a constant, a two-parameter distribution is reduced to a single-parameter distribution. As discussed, for a single-parameter distribution as the consistent distribution and a fixed ϵ , there should be a k -tuple ($d_{im}, \dots, d_{ikm}$) (using a distribution as the consistent distribution means the d values used are calibrated by the distribution or the corresponding kernel distributions generated from this distribution). For a two-parameter distribution, let $D(kurtosis, skewness^2, k, etype, dtype, n) = d_{ikm}$ denote these relations, where the first input is the kurtosis, the second input is the square of the skewness, the third is the order of the central moment (if $k = 1$, the mean), the fourth is the type of estimator, the fifth is the type of consistent distribution, the sixth input is the sample size. Notice that just kurtosis or the square of the skewness is enough to specify the d values for a two-parameter distribution.

Using a two-parameter distribution as the consistent distribution is a problem of robust estimation of parametric models. More specifically, the object is to find solutions for the system of equations

$$\begin{cases} qm(pWA, D(qkurt, qskew^2, 1, quantile, dtype, n)) = \mu \\ qvar(pWvar, D(qkurt, qskew^2, 2, quantile, dtype, n)) = \mu_2 \\ qtm(pWtm, D(qkurt, qskew^2, 3, quantile, dtype, n)) = \mu_3 \\ qfm(pWfm, D(qkurt, qskew^2, 4, quantile, dtype, n)) = \mu_4 \\ qskew = \frac{\mu_3}{\mu_2^2} \\ qkurt = \frac{\mu_4}{\mu_2^2} \end{cases},$$

where μ_2 , μ_3 and μ_4 are the population second, third and fourth central moments. Asymptotically, $qkurt$ and $qskew$ should be fix points of the functions

$$\begin{aligned} \kappa(qkurt, qskew^2) &= \left(\frac{qfm(pWfm, D(qkurt, qskew^2, 4))}{qvar(pWvar, D(qkurt, qskew^2, 2))^2} \right. \\ &\quad \left. \left(\frac{qtm(pWtm, D(qkurt, qskew^2, 3))}{qvar(pWvar, D(qkurt, qskew^2, 2))^{\frac{3}{2}}} \right)^2 \right) \end{aligned}$$

At this stage, D can only be approximated by a Monte Carlo study. Without knowing the details of D , the uniqueness of the fix points is not guaranteed (therefore the asymptotic consistency is not proven). However, it is also clear that this is essentially an overdetermined nonlinear system of equations, because the skewness and kurtosis are interrelated for a two-parameter distribution. Since an overdetermined system is almost always inconsistent when constructed with random coefficients, this becomes a minimization problem in practice. It is still natural to use the fix-point iteration (see Algorithm 1) to find the solutions of

$$\begin{aligned} &\underset{qkurt, qskew^2}{\text{minimize}} \quad |\kappa(qkurt, qskew^2) - (qkurt, qskew^2)| \\ &\text{subject to} \quad (qkurt, qskew^2) \in \text{the domain of } D. \end{aligned}$$

This enables the robust estimations of all four moments to reach a near-consistent level for all common unimodal distributions (Table 1, SI Dataset S1). An advantage compared to M-estimator is that, the optimization here is only related to the function of d value, not the original sample (except for the quantile moments, the re-computation of the quantile function is needed, but the time complexity is $O(1)$ for a sorted sample), so the computational cost is independent of the sample size.

Algorithm 1 $qskew$ and $qkurt$ for a two-parameter distribution

Input: D ; $pWvar$; $pWtm$; $pWfm$; $maxit$; δ

Output: $(qkurt_i, qskew_i^2)$

Check the existence of a solution.

2: $i = 0$

$pair_i \leftarrow \kappa(qkurt = 9, qskew^2 = 4)$ $\triangleright$

Using the d values from the exponential distribution as an initial guess, while other guess can also be specified.

4: **repeat**

$i = i + 1$

6: Given $pair_{i-1}$, use the closest pair of points algorithm to find the closest pair $pair_D$ in the domain of D

$pair_{i-1} \leftarrow pair_D$

8: $pair_i \leftarrow \kappa(pair_{i-1})$

until $i > maxit$ or $|pair_i - pair_{i-1}| < \delta$ $\triangleright maxit$ is the maximum number of iterations, δ is a small positive number. Here, $maxit = 1000$, $\delta = 10^{-6}$.

D. Variance. As the fundamental theorem in statistics, the central limit theorem states that the standard deviation of the limiting form of the sampling distribution of the sample mean is $\frac{\sigma}{\sqrt{n}}$. The principle was later applied to the sampling distributions of the robust location estimators (2, 36–42) and it was found that the efficiencies of the robust location estimators are sometimes very different from the arithmetic mean. Daniell

(1920) stated (36) that the comparison of efficiencies of the various kinds of estimators is useless unless they all tend to coincide asymptotically. Bickel and Lehmann argued, also in the landmark series (40, 41), that meaningful comparisons can be made by studying the standardized variance, asymptotic variances, and sharp lower bounds of these estimators. Here, the scaled standard error (SSE) is proposed to estimate the variances of all estimators, including robust/quantile moments, on a scale similar to that of the sample mean.

Definition D.1 (Scaled standard error). Let $\mathcal{M}_{s_i s_j} \in \mathbb{R}^{i \times j}$ denote the sample-by-statistics matrix, i.e., the first column is the main statistics of interest, $\widehat{\theta}_m$, the second to the j th column are $j - 1$ statistics required to scale, $\widehat{\theta}_{r_1}, \widehat{\theta}_{r_2}, \dots, \widehat{\theta}_{r_{j-1}}$. Then, the scaling factor $\mathcal{S} = \left[1, \frac{\widehat{\theta}_{r_1}}{\widehat{\theta}_m}, \frac{\widehat{\theta}_{r_2}}{\widehat{\theta}_m}, \dots, \frac{\widehat{\theta}_{r_{j-1}}}{\widehat{\theta}_m} \right]^T$ is a $j \times 1$ matrix, which $\bar{\theta}$ is the mean of the column. The normalized matrix is $\mathcal{M}_{s_i s_j}^N = \mathcal{M}_{s_i s_j} \mathcal{S}$. The SSEs are the unbiased standard deviations of the corresponding columns.

Setting the bootstrap moments as the main, the SSEs of all robust estimators proposed here are often between those of the median moments and those of the sample moments (SI Dataset S1). This is because similar monotonic relations between robustness and variance are also very common, e.g., Bickel and Lehmann (41) proved that a low bound for the efficiency of TM_ϵ to sample mean is $(1 - 2\epsilon)^2$ and this monotonic bound is valid for any distribution. Lai, Robbins, and Yu (1983) proposed an estimator that adaptively chooses the mean or median in a symmetric distribution and showed that the result is typically as good as the better of sample mean and median regarding variance (43). It can be seen as an attempt to use the variance version of mean-symmetric weighted average-median inequality. While they used bootstrap standard error as the criterion, another approach can be dated back to Laplace (1812) (44, 45) is using a linear combination of median and mean and the weight is deduced to achieve minimum variance; examples for symmetric distributions see Samuel-Cahn, Chan and He, and Damilano and Puig (46–48).

Scaled standard error enables the direct comparison of variances of different location estimators for asymmetric distributions. Here, two invariant means and related weighted averages ($BM_{\frac{1}{8}}, SQM_{\frac{1}{8}}, WM_{\frac{1}{8}}$, and $BWM_{\frac{1}{8}}$ used here) can create eight possible combinations. Each combination has an SSE for each kurtosis, which can be inferred by a Monte Carlo study. Then, among eight possible combinations, there is one that has the smallest SSE. Theoretically, bootstrap is the optimal way to infer the variance-optimal choice without distributional assumption, however, the computational cost is very high [†]. Similar to Subsection C, let $I(kurtosis, k, dtype, n) = ikm_{WA}$ denote these relations. Then the same kurtosis-specified optimization algorithm can be used to choose the variance-optimum combination (bias-optimum is the same, replacing the SSE by asymptotic bias and then plug in). The only difference is that unlike D , I is discontinuous, so a converged solution might not exist, but setting a proper maximum number of iterations can still get a desired result in practice.

The result is often very close to the optimum choice (SI Dataset S1) if the distributional assumption is correct. In general, compared to the unbiased sample central moments,

[†] The bootstrap-SSE specified optimization will also be included in the package “NRS”, but it is not discussed here, since running all the codes here requires one day, if adding the bootstrap, the testing will need at least one week for a typical PC.

the variances of invariant central moments are much smaller (except the second central moment, Table 1). In 1958, Richtmyer proposed quasi-Monte Carlo simulation based on low-discrepancy sequences, which dramatically reduces the computational cost of large sample simulation (49, 50). Do and Hall extended the principle to bootstrap in 1991 (51) and found that the quasi-random approach is competitive in terms of variance when compared with other bootstrap Monte Carlo procedures. By using quasi-sampling, the impact of the number of repetitions of the bootstrap, or bootstrap size, on variance is negligible. An estimator based on the quasi-bootstrap approach can be seen as a very complex deterministic estimator which is not only computationally efficient, but also statistical efficient. The only drawback is that a small bootstrap size can produce additional finite sample bias but can be corrected by recalibrating the d values (18 thousand is used here as default as it balances computational cost and finite sample bias, except the asymptotic value calculation).

E. Robustness. The measure of robustness used here is the breakdown point proposed by Hampel (52) in 1968. However, the sample-dependent breakdown point has apparently not been formally defined previously.

Definition E.1 (Sample-dependent breakdown point). An estimator $\hat{\theta}$ has a sample-dependent breakdown point if and only if its asymptotic breakdown point $\epsilon(\hat{\theta}, R, \zeta, v)$ is zero and the empirical influence function of $\hat{\theta}$ is bounded, where R is the measure of badness, ζ is the contaminating processes, v is the uncontaminated process. For a full formal definition of the asymptotic breakdown point, which is the breakdown point when $n \rightarrow \infty$, and the empirical influence function, the reader is referred to Genton and Lucas (2003) and Devlin, Gnanadesikan and Kettenring (1975)'s papers (53, 54).

Bear in mind that it differs from the "infinitesimal robustness" defined by Hampel, which is related to whether the asymptotic influence function is bounded (55–57). The proof of the consistency of MoM assumes that it is an estimator with a sample-dependent breakdown point since its breakdown point is $\frac{\beta}{2n}$, where β is the number of blocks, then $\lim_{n \rightarrow \infty} \left(\frac{\beta}{2n}\right) = 0$, if β is a constant or $\beta \ll n$, and any changes in any one of the points of the sample cannot breakdown this estimator (58–60).

For the robust estimations of central moments or other robustified U -statistics based on a robust location estimator, the asymptotic breakdown points are suggested by the following theorem by extending the method in Donoho and Huber (1983)'s proof of the breakdown point of the Hodges-Lehmann estimator (61).

Theorem E.1. *Given n independent random variables $(X_1, \dots, X_n)$ with the same distribution F and a U -statistic (first generalized by Hoeffding in 1948) (62) associated with a symmetric kernel of degree k . Then, assuming as $n \rightarrow \infty$, $k \ll n$, the asymptotic breakdown point (52, 53, 55, 57, 61) of the robust location estimation of the distribution of the kernel of the U -statistic is $1 - (1 - \epsilon)^{\frac{1}{k}}$, where ϵ is the breakdown point of the symmetric robust location estimator.*

Proof. According to the definition of ϵ -contamination (61), suppose m contaminants are added to the sample. The fraction of bad values in the sample is $\epsilon_U = \frac{m}{n+m}$, while the original n data points are not impacted. That means, in the distribution of the kernel, $\binom{n}{k}$ of total $\binom{n+m}{k}$ points are not corrupted.

Then, the breakdown will not occur if the following inequality holds

$$\binom{n}{k} > \left(\frac{1}{\epsilon} - 1\right) \times \left(\binom{n+m}{k} - \binom{n}{k}\right),$$

Since ϵ is the breakdown point of the robust location estimator, $\frac{1}{2} \geq \epsilon \geq 0$,

$$\frac{1}{1-\epsilon} > \frac{\binom{n+m}{k}}{\binom{n}{k}} = \frac{(n+m)(n+m-1)\dots(n+m-k+1)}{n(n-1)\dots(n-k+1)}.$$

For asymptotic breakdown point, assuming $n \rightarrow \infty$, $k \ll n$, $\lim_{n \rightarrow \infty} \left(\frac{n+m-k+1}{n-k+1}\right) = \frac{n+m}{n} = x$, then

the above inequality does not hold when $x \geq \left(\frac{1}{1-\epsilon}\right)^{\frac{1}{k}}$.

So, the breakdown point of the U -statistic is $\epsilon_U = \frac{m}{n+m} = 1 - \frac{n}{n+m} = 1 - \frac{1}{x} = 1 - (1 - \epsilon)^{\frac{1}{k}}$. $\square$

Remark. If $k = 1$, $1 - (1 - \epsilon)^{\frac{1}{k}} = \epsilon$, so this formula also holds for the robust location estimator. In addition, the numerical solutions for $k = 2, 3, 4$, $\epsilon = \frac{1}{8}$ are ≈ 0.065 , 0.044 , and 0.033 , respectively. When $\epsilon = \frac{1}{2}$, the robustified U -statistic becomes U -quantile, which converges almost surely as proven by Choudhury and Serfling (63) in 1988.

Discussion

Moments, including raw moments, central moments, and standardized moments, are key parameters that determine probability distributions. The central moments are much more popular than the raw moments because they are invariant to translation. In 1947, Hsu and Robbins proved that the arithmetic mean converges completely to the population mean provided the second moment is finite (64). The strong law of large numbers implies that the k th sample central moment is asymptotically unbiased. Parametrics, semi-parametrics, and non-parametrics are the current three main branches of statistics. Maximum likelihood was first proposed by Fisher in 1922 (65) in a multinomial model and then generalized by Cramér (1946), Hájek (1970), and Le Cam (1972) (25, 39, 66). Besides Newcomb (2, 3), the general robust estimation of parametric models dates back to 1939, when Wald (67) suggested the use of minimax estimates to solve such problems. Hodges and Lehmann in 1950 (68) expanded upon this concept and obtained minimax estimates for a series of important problems. It was soon clear that a minimax estimator should be a Bayes estimator with regard to the least favorable prior distribution of θ since a minimax estimator is the best in the worst case. The adaptive estimation of the center of symmetry was studied by van Eeden (1970) and Takeuchi (1971) (69, 70). Bickel, in 1982, simplified the general heuristic necessary condition proposed by Stein (71) (1956) and derived sufficient conditions for this type of problem, adaptive estimation (72). As pointed out in Begun, Hall, Huang, and Wellner's paper (1983) and their semiparametric book with Bickel, Klaassen, and Ritov (1993) (73, 74), the two problems, semiparametrics (or adaptive estimation) and parametric robustness, are closely related but different. The paradigm shift here opens up a possibility that by defining a semi-parametric model with inequalities, constructing estimators simultaneously for two or more very different semi-parametric/parametric models within the semi-parametric model, then even for a seemingly "wrong"

parametric model that belongs to the semi-parametric model but not belonging to the parametric models used to calibrate, the performance might still be near-optimal owing to the common nature of the models used by the estimators. Also, the models can be directly expanded and not limited to a single parametric form. Maybe it can be named as comparametrics. Closely related topics are "mixture model" and "constraint defined model" generalized in (74) and the method of sieves, introduced by Grenander (75).

Recently, fascinating statistical phenomena about Taylor's law for distributions with infinite mean were found by Drton and Xiao (2016) (76), Pillai and Meng (2016) (77), Cohen, Davis, and Samorodnitsky (2020) (78), and Brown, Cohen, Tang, and Yam (2021) (79). Lindquist and Rachev (2021) commented: "What are the proper measures for the location, spread, asymmetry, and dependence (association) for random samples with infinite mean?" (80). This is not the focus of this paper, but it is almost sure that the estimators proposed here will have a place. For example, they (80) suggested using median, interquartile range, and medcouple (81) as the robust versions of these moments (82–84). Obviously now, if one wants to preserve the original relation between each moment and at the same time wants the measures to be as robust as possible, the natural choices should be median, median variance, and median third central moment. Analogous to trimmed L-moment (85), which is the most robust version of L-moment (86), the central moment now also has its standard most robust version. As the building blocks of statistics, NRSs provide an option for estimating population moments robustly and near-consistently with moderate variances under mild assumptions, thus improving the consistency of statistical results across studies, especially when heavy-tailed distributions may be present (87–89).

Methods

This section describes the methods for generating Table 1.

The computation of average bias was the same as that of average asymptotic bias as described previously. Here, $\delta = 0.1$, $C = 120$. For the Weibull, gamma, lognormal and generalized Gaussian distributions, the kurtosis range is from 3 to 15 (there are two shape parameter solutions for the Weibull distribution, the lower one is used here). For the Pareto distribution, the range is from 9 to 21.

Standardization is crucial for comparison among different distributions. Currently, there are many choices, e.g., Gauss used the root mean square deviation from the mode, whereas mean absolute deviation and standard deviation are two common choices. Standard deviation is used here because of its central role in standard error estimation. The estimation of central moments based on the location estimations of the kernel distributions also enables the standardization of ABs and AABs of invariant central moments. The only difference is that the population standard deviation is replaced by the standard deviation of the kernel distribution (σ_{km}).

Then, analogous to the asymptotic bias, the scaled standard error can be standardized, averaged, and weighted (weighted scaled standard error is also defined as weighting the results of the average scaled standard error with the corresponding numbers of Google Scholar search results). It should be noted that in Table 1, for symmetric distributions, generalized Gaussian, the standard errors are used for location and asymmetry estimators, since when the mean value is close to zero, the scaled standard error will approach infinity and is therefore too sensitive to small changes.

Asymptotic d values for the invariant central moments for the exponential distribution were approximated by a quasi-Monte Carlo study (49, 50) based on generating a large quasi-random sample with sample size 1.8 million from an exponential distribution and quasi-subsampling the sample 1.8 million times to approximate

the distributions of the kernels of the corresponding U -statistics, then computing the binomial k th central moment (Bkm), median k th central moment (mkm), and corresponding quantiles, finally obtained by the formula $d_{rkm} = \frac{km_{bs} - Bkm_{bs}}{Bkm_{bs} - mkm_{bs}}$ and

$$d_{qkm} = \frac{pkm_{bs} - pBkm_{bs}}{pBkm_{bs} - \frac{1}{2}}, \text{ where } pBkm = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{X_i \leq (Bkm_{bs})},$$

$$pkm = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{X_i \leq (km_{bs})}, \text{ } bs \text{ indicates the bootstrap moments.}$$

The accuracies were checked by comparing the bootstrap central moments to the asymptotic values (errors), showing that for the second, third, and fourth central moments, the errors are ≈ 0.0003 , ≈ 0.001 , and ≈ 0.03 , respectively. The sample standard deviation of the kernel distributions of the second, third, and fourth central moments are 2.234, 9.627, and 60.064, respectively, so, the errors of bias after standardization for the second, third, and fourth central moments were all smaller than 0.001, thus ensuring the accuracy implied in the number of significant digits of the values in Table 1. Invariant moments using other weighted averages, $SQM_{\frac{1}{8}}$, $WM_{\frac{1}{8}}$, and $BWM_{\frac{1}{8}}$, are the same, just plugging in. For finite sample

size, $n = 5400$, the d values were calculated by using the Monte Carlo results (2000 pseudorandom samples) with bootstrap size 18 thousands. For the errors of d value estimations, the first order Taylor approximation of the d value function is $d = \frac{x_1 - x_2}{x_2 - x_3} \approx$

$$d^0 + \frac{\partial d}{\partial x_1} x_1 + \frac{\partial d}{\partial x_2} x_2 + \frac{\partial d}{\partial x_3} x_3. \text{ Then, using the Bienaymé's identity,}$$

$$\text{the variance of } d \text{ can be approximated by } \sigma_d^2 \approx \left| \frac{\partial d}{\partial x_1} \right|^2 \sigma_{x_1}^2 +$$

$$\left| \frac{\partial d}{\partial x_2} \right|^2 \sigma_{x_2}^2 + \left| \frac{\partial d}{\partial x_3} \right|^2 \sigma_{x_3}^2 + 2 \left| \frac{\partial d}{\partial x_1} \right| \left| \frac{\partial d}{\partial x_2} \right| Cov(X_1, X_2) +$$

$$2 \left| \frac{\partial d}{\partial x_1} \right| \left| \frac{\partial d}{\partial x_3} \right| Cov(X_1, X_3) + 2 \left| \frac{\partial d}{\partial x_2} \right| \left| \frac{\partial d}{\partial x_3} \right| Cov(X_2, X_3) =$$

$$\left| \frac{1}{x_2 - x_3} \right|^2 \sigma_{x_1}^2 + \left| -\frac{x_1 - x_2}{(x_2 - x_3)^2} - \frac{1}{x_2 - x_3} \right|^2 \sigma_{x_2}^2 + \left| \frac{x_1 - x_2}{(x_2 - x_3)^2} \right|^2 \sigma_{x_3}^2 +$$

$$2 \left| \frac{1}{x_2 - x_3} \right| \left| -\frac{x_1 - x_2}{(x_2 - x_3)^2} - \frac{1}{x_2 - x_3} \right| Cov(X_1, X_2) +$$

$$2 \left| \frac{1}{x_2 - x_3} \right| \left| \frac{x_1 - x_2}{(x_2 - x_3)^2} \right| Cov(X_1, X_3) +$$

$$2 \left| -\frac{x_1 - x_2}{(x_2 - x_3)^2} - \frac{1}{x_2 - x_3} \right| \left| \frac{x_1 - x_2}{(x_2 - x_3)^2} \right| Cov(X_2, X_3). \text{ Since}$$

$$\text{for the robust mean, } \sigma_{x_1}^2 = 0, \text{ so, } \sigma_{d_{rm}}^2 \approx$$

$$\left| -\frac{x_1 - x_2}{(x_2 - x_3)^2} - \frac{1}{x_2 - x_3} \right|^2 \sigma_{x_2}^2 + \left| \frac{x_1 - x_2}{(x_2 - x_3)^2} \right|^2 \sigma_{x_3}^2 +$$

$$2 \left(-\frac{x_1 - x_2}{(x_2 - x_3)^2} - \frac{1}{x_2 - x_3} \right) \left(\frac{x_1 - x_2}{(x_2 - x_3)^2} \right) Cov(X_2, X_3), \text{ where}$$

$$x_1 \text{ is the expected value, } x_2 \text{ is the weighted average used,}$$

$$x_3 \text{ is the median. For quantile mean, since } \sigma_{x_3}^2 = 0,$$

$$\sigma_{d_{qm}}^2 \approx \left| \frac{1}{x_2 - x_3} \right|^2 \sigma_{x_1}^2 + \left| -\frac{x_1 - x_2}{(x_2 - x_3)^2} - \frac{1}{x_2 - x_3} \right|^2 \sigma_{x_2}^2 +$$

$$2 \left(\frac{1}{x_2 - x_3} \right) \left(-\frac{x_1 - x_2}{(x_2 - x_3)^2} - \frac{1}{x_2 - x_3} \right) Cov(X_1, X_2), \text{ where } x_1 \text{ is}$$

$$\text{the percentile of the expected value, } x_2 \text{ is the percentile of the}$$

$$\text{weighted average used, } x_3 \text{ is the percentile of median, } \frac{1}{2}. \text{ The}$$

$$\text{errors of } d \text{ values were estimated by the corresponding sample}$$

$$\text{statistics. The computations of ABs and AABs for invariant central}$$

$$\text{moments were just the same as mentioned previously. The SSE}$$

$$\text{was computed by approximating the sampling distribution with}$$

$$2000 \text{ pseudorandom samples for } n = 5400 \text{ and } 30 \text{ pseudorandom}$$

$$\text{samples for } n = 1.8 \times 10^6. \text{ Common random numbers were}$$

$$\text{used for better comparison. The errors of AB and SSE were}$$

$$\text{estimated by } se(\bar{x}) = \frac{\sigma}{\sqrt{n}} \approx \frac{usb}{\sqrt{n}}, \text{ } se(sd) \approx \frac{1}{2\sigma} se(var) =$$

$$\sqrt{\frac{\mu_4}{4n\sigma^2} - \frac{n-3}{4n(n-1)}\sigma^2} \approx \sqrt{\frac{fm}{4nvar} - \frac{n-3}{4n(n-1)}var}, \text{ where } usb \text{ is}$$

$$\text{unbiased standard deviation of the sampling distribution with}$$

$$\text{normality assumption (90). The simulation results and related}$$

$$\text{codes are all included in the SI Dataset S1 and in GitHub.}$$

Data Availability. Data for Table 1 are given in SI Dataset S1.

All codes have been deposited in GitHub.

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